Energy Disorder & Obesity: Saturated Fatty Acid Induced Hypothalamic Neurodegeneration

Gabe A. De Los Rios, Cayla M. Duffy, Kathryn Pawley, Tammy A. Butterick, & ChuanFeng Wang
Department of Food Science and Nutrition, University of Minnesota

Introduction

High fat diet (HFD), particularly diets high in saturated fats, are associated with increase in weight gain resulting in obesity. Palmitic acid (PA), one of the main constituents of saturated fatty acids in HFD, has been shown to increase insulin resistance in the hypothalamus of rodents.[1] HFD is also thought to increase levels of ceramides, fatty acid metabolites that have been shown to cause apoptosis, or programmed cell death, in neurons.[2, 3] Neurodegeneration of the hypothalamus, the center of the brain responsible for certain metabolic process, could contribute to many adverse effects such as adult-onset obesity.[4]

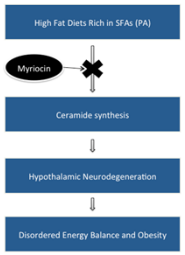


Figure 1. A schematic overview of the central hypothesis. Myriocin inhibits the SPT-1 enzyme needed to synthesize ceramides from high fat diets rich in saturated fatty acids (SFA).

Myriocin (myr), a potential ceramide inhibitor, has been shown to inhibit serine palmitoyl transferase-1 (SPT1), the rate-limiting enzyme of *de novo* ceramide synthesis[5] (Fig. 1, 4). **We hypothesize that palmitic acid induces hypothalamic neurodegeneration, mediated by the *de novo* synthesis of ceramides.** We aimed to test this hypothesis in a hypothalamic cell line treated with PA. We predicted that myriocin would interrupt *de novo* ceramide synthesis following PA treatment, thus preventing neurodegeneration.

Results

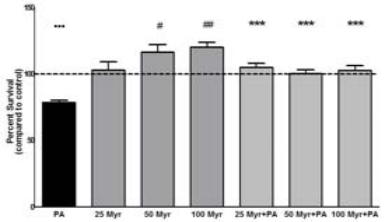


Figure 2. After 24 h pretreatment with myriocin (myr; 25, 50, or 100 nmol/µL), cells were challenged with palmitic acid (PA; 0.1 mM), treated with a second dose of myr, and incubated for 24 h. Cell viability expressed as relative fluorescent units normalized to control. Statistical differences were determined by an unpaired two tailed t-test (… $p < 0.0001$ and ## $p = 0.0019$ vs. control; *** $p < 0.0001$ vs. PA alone)

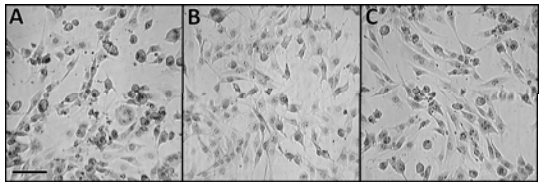


Figure 3. Hypothalamic cells after 24 h pretreatment with 0.1 mM palmitic acid alone (A), 100 nmol/µl myriocin (B), or palmitic acid plus myriocin (C). Cells shown here are representative replicates of the cells used for the viability assay data shown in Fig. 2 above. Image taken at 10x magnification. Scale bar = 100 µm.

- **Palmitic acid (PA) does decrease cell viability**
- **Myriocin (25, 50, and 100 nmol) dose-dependently attenuates PA-induced neurodegeneration**

Methods

Cell Growth and Treatment
Immortalized primary mouse hypothalamic neurons (A1/2 cell line) (CELLutions Biosystems Inc) were used for this experiment. Cells were grown in complete media (1x DMEM (Sigma) supplemented with 10% FBS (PAA labs) and 1% PSN (PSN mix, Invitrogen)) in 75 cm^2 vented flasks in a humidified incubator at 37°C with 5% CO_2.
For each treatment, cells were plated (~3500 cells/well) into a 96 well plate. Cells were pretreated for 24 h with increasing concentrations of myriocin (Sigma; 25, 50, and 100 nmol). Cells were then challenged with 50 µl of 0.1 mM PA , given a second dose of myriocin at concentrations listed above, and incubated for a further 24 hours.
Assays
Cell viability was measured using 10% PrestoBlue™ Cell Viability Reagent (Invitrogen)[6]. After 10 minutes of incubation at optimal growth conditions, 96 well plates were measured for relative fluorescence units (RFUs) at excitation wavelength of 560 nm and emission wavelength of 590 nm using a SpectraMax M5 multi-detection reader (Molecular Devices).
Data Analysis
Data analysis was performed using GraphPad Prism 5 utilizing two-tailed unpaired T-test to determine differences between vehicle control and treatment groups. Prior to analysis all treatment groups were normalized to vehicle control.

Acknowledgements

I would like to give a special thanks to TRiO McNair staff as well as the MnONLab. I would also like to acknowledge the following people for all their support and help: Drs. Tammy Butterick & Joshua Nixon

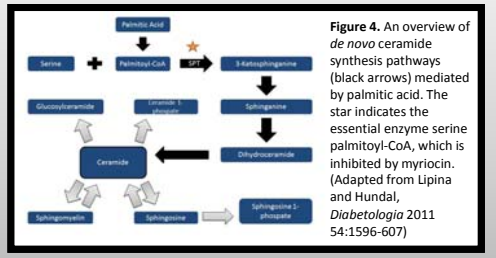


Figure 4. An overview of *de novo* ceramide synthesis pathways (black arrows) mediated by palmitic acid. The star indicates the essential enzyme serine palmitoyl-CoA, which is inhibited by myriocin. (Adapted from Lipina and Hundal, *Diabetologia* 2011 54:1596-607)

Discussion

As myriocin is a known SPT1 inhibitor, the significantly higher level of cell viability in cells treated with myr and PA suggest that the neurodegeneration induced through the ceramide synthesis *de novo* pathway rather than directly caused by PA itself. In addition, our results showed myr-treated cells had significantly greater viability compared to control, suggesting that myriocin may promote neuronal cell survival. Studies are ongoing to determine this observation.

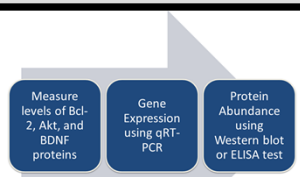


Figure 5. Schematic outline of future experiments to further study the neuroprotective and/or neurodegenerative properties of myriocin and palmitic acid.

To further evaluate the neurodegenerative effect of PA, gene expression levels of B cell leukemia 2 protein (Bcl-2), brain-derived neurotrophic factor (BDNF), and protein kinase B (Akt) will be measured in cells challenged with a combination of PA and myr. Bcl-2 has been shown to be an anti-apoptotic mediator in high fat diets.[7] In addition to this, Bcl-2 levels are also increased by higher levels of BDNF, a neurotrophin involved in neuroprotection, neuronal survival, differentiation and energy regulation.[8]

References

1. Benoit, S.C., et al., *Palmitic acid mediates hypothalamic insulin resistance by altering PKC-theta subcellular localization in rodents.* The Journal of clinical investigation, 2009. **119**(9): p. 2577-89.
2. Blachnio-Zabielska, A., et al., *Effect of high fat diet enriched with unsaturated and diet rich in saturated fatty acids on sphingolipid metabolism in rat skeletal muscle.* J Cell Physiol, 2010. **225**(5): p. 786-91.
3. Tong, M. and S.M. de la Monte, *Mechanisms of ceramide-mediated neurodegeneration.* J Alzheimers Dis, 2009. **16**(4): p. 705-14.
4. Ryu, K.Y., et al., *Hypothalamic neurodegeneration and adult-onset obesity in mice lacking the Ubb polyubiquitin gene.* Proc Natl Acad Sci U S A, 2008. **105**(10): p. 4016-21.
5. Usher, J.R., et al., *Inhibition of de novo ceramide synthesis reverses diet-induced insulin resistance and enhances whole-body oxygen consumption.* Diabetes, 2010. **59**(10): p. 2453-64.
6. Xiao, J., et al., *Monitoring of cell viability and proliferation in hydrogel-encapsulated system by resazurin assay.* Appl Biochem Biotechnol, 2010. **162**(7): p. 1996-2007.
7. Moraes, J.C., et al., *High-fat diet induces apoptosis of hypothalamic neurons.* PLoS One, 2009. **4**(4): p. e5045.
8. Almeida, R.D., et al., *Neuroprotection by BDNF against glutamate-induced apoptotic cell death is mediated by ERK and PI3-kinase pathways.* Cell Death Differ, 2005. **12**(10): p. 1329-43.